

REPORT ON

PLC-Based Petrol Pump Control System

COURSE TITLE:-

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INTELLIGENT INDUSTRIAL AUTOMATION: AI, SENSORS, ACTUATORS AND PLC INTEGRATION

1. About the Topic

A Programmable Logic Controller (PLC) is an industrial digital controller used to automate machines and industrial processes. This project simulates a PLC-based Petrol Pump Control System using LogixPro. The system allows the operator to select the fuel type (Petrol or Diesel), choose the required quantity (1 L, 2 L, or 5 L), and dispense fuel safely using ladder logic. Safety features such as an Emergency Stop and Overflow Sensor are included to improve system reliability.

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3. Introduction to the Project

The PLC-Based Petrol Pump Control System is a simulation developed in LogixPro to demonstrate industrial automation using ladder logic. The system allows selection of fuel type and dispensing quantity, controls the pump using timers, prevents invalid selections through XOR logic, and improves safety using an Emergency Stop and Overflow Sensor. A customer counter records the number of completed dispensing operations.

4. Applications

- Petrol filling stations
- Diesel dispensing stations
- Fuel dispensing terminals
- Industrial automation training
- PLC laboratory demonstrations
- Educational simulation of fuel dispensing systems

5. Ladder Diagram Description

Rung 000 – Main Pump Control

This is the main control rung of the petrol pump. It validates Petrol/Diesel selection using XOR logic, validates the selected quantity (1 L, 2 L, or 5 L), checks the START push button, Emergency Stop, and Overflow Sensor, and energizes output O:2/0 (Pump Motor). The O:2/0 contact is connected in parallel with the START push button to create a seal-in (holding) circuit, allowing the pump to continue running after START is released.

Rung 001 – 1 L Timer

When 1 L is selected, the pump is running, and the Emergency Stop is inactive, TON Timer T4:0 starts. The timer preset is 100 (10 seconds).

Rung 002 – 2 L Timer

When 2 L is selected, the pump is running, and the Emergency Stop is inactive, TON Timer T4:1 starts. The timer preset is 200 (20 seconds).

Rung 003 – 5 L Timer

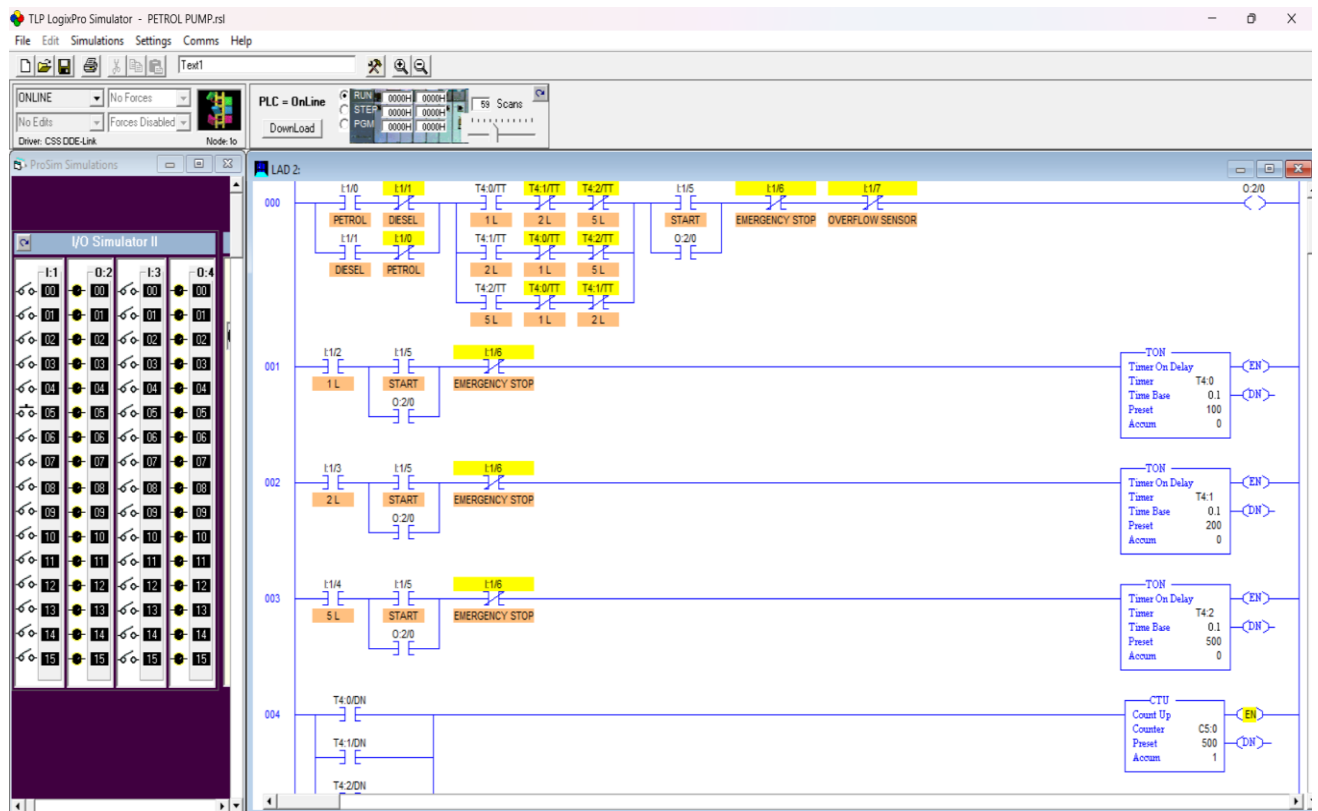
When 5 L is selected, the pump is running, and the Emergency Stop is inactive, TON Timer T4:2 starts. The timer preset is 500 (50 seconds).

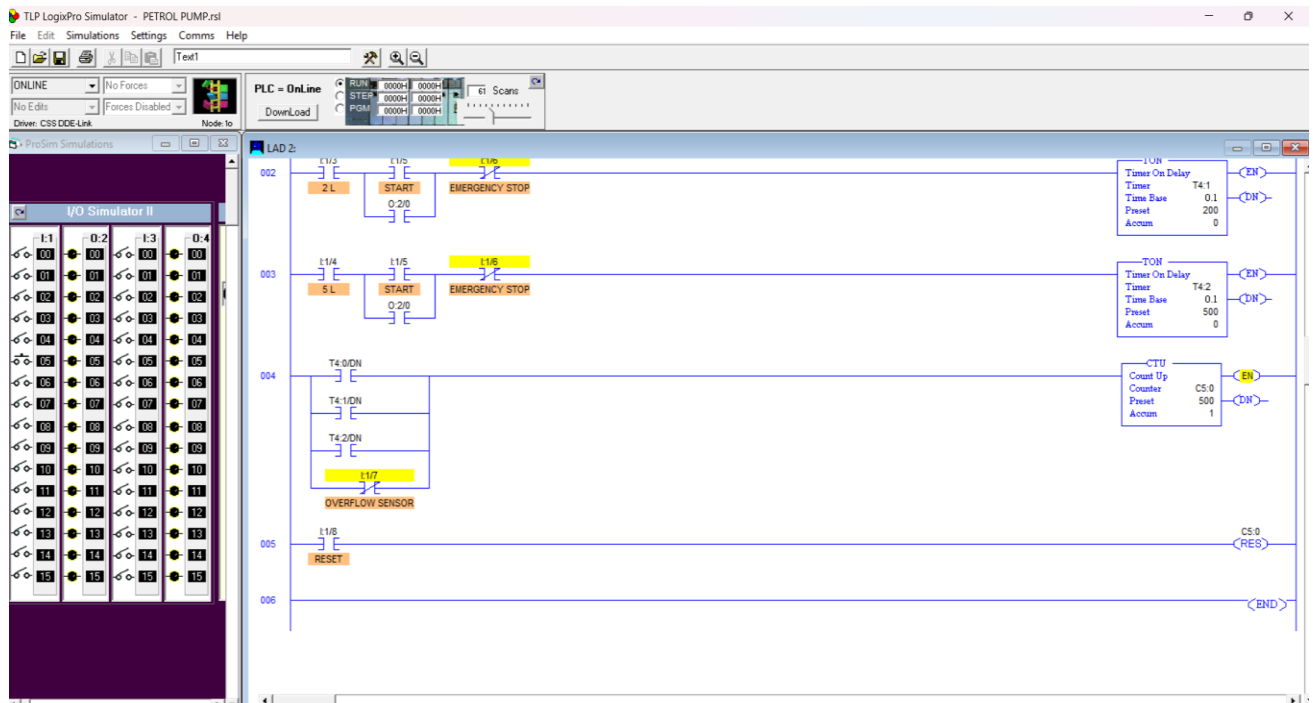
Rung 004 – Customer Counter

The CTU counter increments whenever T4:0/DN, T4:1/DN, or T4:2/DN becomes true, indicating successful completion of dispensing.

Rung 005 – Counter Reset

Pressing the RESET push button executes the RES instruction and resets Counter C5:0 to zero.





6. Output

The developed ladder logic successfully simulates a petrol pump control system. The program validates fuel and quantity selection, starts the pump, controls dispensing using timers, stops operation during emergency or overflow conditions, and counts successfully completed customer transactions.

7. Advantages

- Simple and reliable PLC-based automation
- Prevents invalid fuel and quantity selections
- Improves operational safety using Emergency Stop and Overflow Sensor
- Automatic customer counting
- Easy to modify and expand

8. Disadvantages

- Simulation only; no real fuel flow measurement
- No price calculation or billing
- Manual input switch operation in the simulator
- Does not include HMI or display panel

9. Working

The operator selects either Petrol or Diesel and then chooses one dispensing quantity (1 L, 2 L, or 5 L). XOR logic ensures that only one fuel type and one quantity are selected. After pressing START, the pump output is energized and the corresponding TON timer begins counting. Once the timer reaches its preset value, the dispensing cycle is completed and the customer counter increments. If the Emergency Stop or Overflow Sensor is activated, the pump stops immediately.

10. Challenges

- Implementing XOR logic for fuel selection
- Implementing XOR logic for quantity selection
- Designing the pump holding circuit
- Selecting appropriate timer values
- Adding overflow protection without affecting customer counting
- Understanding timer and counter behavior during testing

11. Conclusion

The PLC-Based Petrol Pump Control System was successfully implemented using LogixPro ladder logic. The project demonstrates the practical use of PLC timers, counters, selection logic, and safety interlocks in an industrial-style application. It provides a clear understanding of PLC programming concepts and their application in automated fuel dispensing systems.